

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 50

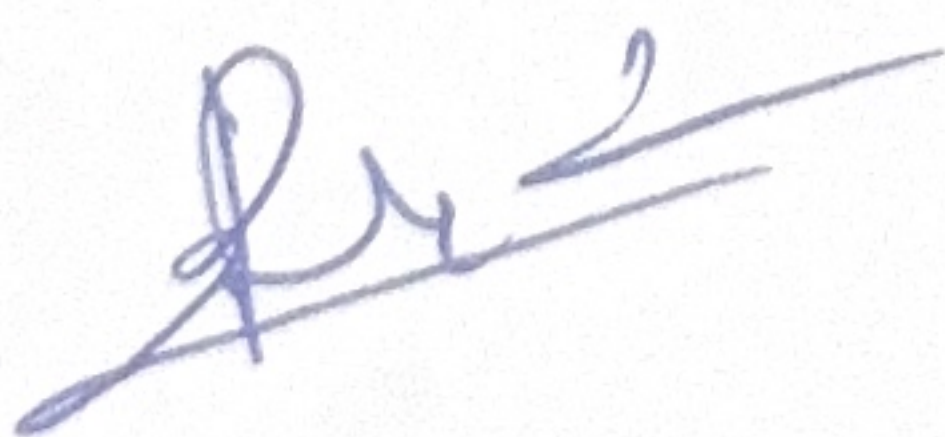
Annexure A

Scenario Based Assignment

Code of Conduct:

I M. PRASANNA VENKATEGH (192573016) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



DISASTER - RESPONSE PRONE

i) GREEDY BEST-FIRST SEARCH:

Greedy Best First Search chooses the node

that appears closest to the destination.

Formula : $f(n) = h(n)$

here; $h(n)$ can be the straight-line distance to the village.

ADVANTAGES:

- * Fast decision making.
- * Simple to implement.
- * Useful in emergencies.

DISADVANTAGES:

- * Ignores the cost already travelled.
- * Does not guarantee the optimal route.
- * Poor heuristic can give bad results.

ii) A* SEARCH:

A* uses both actual path cost and estimated remaining cost. Formula : $f(n) = g(n) + h(n)$. The drone can include distance, energy, terrain and smoke in $g(n)$. A* gives a better balance between cost, distance and safety.

iii) EFFECT OF HEURISTIC ENERGY ACCURACY:

The heuristic strongly affects path algorithms.

Greedy Best-First:

- * Uses only $h(n)$.
- * Very dependent on heuristic accuracy.
- * Wrong estimates can produce poor routes.

A*:

- * Uses $g(n) + h(n)$.
- * Actual path cost reduces dependence on the heuristic.
- * A good heuristic makes A* faster.

ii) DYNAMIC CHANGES:

~~Wildfires~~ can suddenly block a trail.
The drone must:

1. Detect the blocked trail.
2. Remove it from the accessible paths.
3. Find an alternative route.
4. Continue from its current location.

Dynamic environments require continuous monitoring and replanning.

RECOMMENDED ALGORITHM:

A* with dynamic replanning is the best choice.

- * Considers actual travel cost.
- * Uses heuristic guidance.
- * Can consider smoke and terrorism risk.
- * Provides better route quality.

Greedy Best-First is faster but may select unsafe routes.
A* with dynamic replanning provides the best balance of speed, optimality and safety.

SMART IRRIGATION SYSTEM:

Formulation as an Optimization Problem. Each valve setting is a decision variable:

$$X = \{x_1, x_2, \dots, x_3\}$$

The objective is to minimize.

Water Wastage + Energy Consumption + Crop Stress

i) HILL CLIMBING:

Hill climbing starts with one value configuration and changes it to a better neighboring configuration.

ADVANTAGES: Fast, simple and low memory.

LIMITATIONS: Can get stuck in local optima, plateaus and ridges.

CONCLUSION: Useful for quick optimisation but not effective.

ii) LOCAL MAXIMA, PLATEAUS and RIDGES

LOCAL OPTIMUM: A solution better than nearby solutions but not the best overall.

PLATEAU: Many solutions have almost the same value.

RIDGE: Several values must change together to improve the solution.

iii) SIMULATED ANNEALING AND GENETIC ALGORITHM:

SIMULATED ANNEALING:

- * Sometimes accepts a worse solution.
- * Helps escape local optima.
- * Later focuses on better solutions.

GENETIC ALGORITHMS:

- * Uses many solutions.
- * Selects good solutions.
- * Uses crossover and mutation.
- * Finds different possible solutions.

CONCLUSION:

Genetic Algorithm gives good scalability and adaptability.

iv) BEST APPROACH:

Genetic Algorithm is

- * Handles a huge search space
- * Searches many solutions
- * Avoids local optima
- * Can use new information

CONCLUSIONS:

adapt.

3)

BEST APPROACH:

Genetic Algorithm is suitable because:

- * Handles a huge search space.
- * Searches many solutions.
- * Avoids local optima.
- * Can use new sensor data.

CONCLUSIONS:

Genetic Algorithm gives good scalability and adaptability.

3) AUTONOMOUS SUBMARINE

i) ONLINE SEARCH AGENT:

The submarine explores an unknown trench, so it does not have a complete map.

It must: Sense $\rightarrow$ Plan $\rightarrow$ Learn $\rightarrow$ Replan.

Online search allows the submarine to make decisions while exploring.

CONCLUSION: Online search is better than offline search for an unknown environment.

ii) PARTIAL OBSERVABILITY AND DYNAMIC ENVIRONMENT:

The submarine has limited sensor, so it cannot see everything.

It may not know.

* Rocks * Thermal vents * Safe Paths

New hazards may also appear. It must continuously sense and update its route.

ii) INTERNAL MODEL:

The submarine builds a map using sonar.

STEPS:

1. Detect objects.
2. Identify safe and dangerous Areas.
3. Add them to the map.
4. Give Cost to dangerous areas
5. Plan a route.
6. Update the map when new information arrives

BEST APPROACH:

The best approach is Online Informed Search.

It can:

- * Use sonar information.
- * Update the map.
- * Avoid hazards.
- * Use heuristics.
- * Replan when conditions change.

A* can also be used with

$$f(n) = g(n) + h(n).$$

RUBRICS FOR

Criteria	Weightage	Level 4 - Example
Understanding of Case	25	Thorough understanding
Application of Concepts		

Problem-solving techniques and
Planning with heuristic function

Tool Weightage: 40%

QUESTIONS: 3

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

22 + 22 + 18 + 18
 + 9
 (89)